Keywords: multimodal learning • tensor decomposition • synergy • contrastive learning

HRIL: Learning Multimodal Synergy via Higher-Order Tensor Modeling

Tucker Decomposition of a Cross-Moment Tensor Captures the Emergent Information That Pairwise Contrastive Objectives Miss

By Qun Dai, Zhao Kang et al.

arXiv:2610.12393 • NeurIPS 2026 • October 2026

Multimodal contrastive learning fails to capture *synergistic* information—the signal that is only present in the joint combination of three or more modalities. HRIL introduces a cross-moment tensor over modality embeddings and factorises it with Tucker decomposition inside a contrastive objective, with a regulariser that prevents the core tensor from collapsing to pairwise terms.

AT A GLANCE

Problem: Pairwise contrastive losses miss synergistic information that only exists in the joint combination of three or more modalities.

Why it matters: Many real-world multimodal tasks rely on cross-modal synergy; ignoring it leaves performance on the table.

Method: Outer-product cross-moment tensor over modality embeddings, compressed by Tucker decomposition, with a diagonal-suppressing regulariser.

Result: Consistent gains over multimodal contrastive baselines on synergy-dominated benchmarks; accepted to NeurIPS 2026.

Evidence: Ablation confirms the synergy regulariser is necessary; Tucker rank controls the expressiveness–cost trade-off.

The Big Picture

The information-theoretic decomposition of multimodal data identifies three components: *redundant* (shared across modalities), *unique* (present in one modality only), and *synergistic* (present only in the joint combination). Standard contrastive objectives such as InfoNCE maximise pairwise mutual information, capturing redundant and unique information but systematically missing synergy.

HRIL (Higher-order Representation Interaction Learning) addresses this gap by constructing a tensor representation that explicitly encodes interactions of all orders up to M for M modalities, then compressing it efficiently via Tucker decomposition. The result is a contrastive framework that is provably sensitive to synergistic effects.

Why Existing Approaches Fall Short

Pairwise contrastive losses (InfoNCE, SimCLR) consider only two modalities at a time, so terms requiring all M modalities simultaneously are never optimised.

Late fusion combines independent unimodal embeddings at inference time, encoding no interactions before the final decision.

Bilinear tensor fusion captures pairwise interactions and scales quadratically with the number of modalities, missing higher-order terms and becoming intractable for large M.

Core Mechanism

HRIL forms the outer product of modality embeddings:

$$\mathcal{T}^{(i)} = \bigotimes_{m=1}^{M} \mathbf{z}_m^{(i)}, \quad \mathcal{T}^{(i)} \in \mathbb{R}^{d_1 \times \cdots \times d_M}$$

and compresses it via Tucker decomposition into a core tensor $\mathcal{G} \in \mathbb{R}^{r_1 \times \cdots \times r_M}$ with factor matrices $\mathbf{U}_m \in \mathbb{R}^{d_m \times r_m}$. A regulariser penalises the diagonal entries of $\mathcal{G}$ (which correspond to unimodal and pairwise terms), forcing the model to allocate capacity to genuinely multi-way interactions.

Technical Formulation

Let $f_m : \mathcal{X}_m \to \mathbb{R}^{d_m}$ be the encoder for modality m, with output $\mathbf{z}_m^{(i)} = f_m(x_m^{(i)})$.

Tucker factorisation approximates the cross-moment tensor:

$$\hat{\mathcal{T}} = \mathcal{G} \times_1 \mathbf{U}_1 \cdots \times_M \mathbf{U}_M, \quad \|\mathcal{T} - \hat{\mathcal{T}}\|_F \leq \epsilon$$

The synergy regulariser penalises diagonal concentration:

$$\mathcal{L}_{\text{syn}} = \sum_r \mathcal{G}^2_{r,r,\ldots,r}$$

The total objective balances contrastive alignment and synergy:

$$\mathcal{L} = \mathcal{L}_{\text{InfoNCE}} + \lambda \mathcal{L}_{\text{syn}}$$

where $\lambda > 0$ is a hyperparameter.

Learning or Inference Procedure

Training:

1. Encode each modality: $\mathbf{z}_m^{(i)} = f_m(x_m^{(i)})$.
2. Form the outer-product tensor $\mathcal{T}^{(i)}$.

3. Project through Tucker factors to obtain the core $\mathcal{G}^{(i)}$.
4. Evaluate the contrastive loss over positive/negative pairs.
5. Add $\lambda\mathcal{L}_{\text{syn}}$ and backpropagate through Tucker factors and encoders jointly.

Inference: The Tucker-compressed representation provides a fused embedding for downstream classification or retrieval.

What the Guarantee Says

Tucker decomposition provides a controlled approximation: the Frobenius reconstruction error is bounded by the residual from truncating the Tucker rank. The synergy regulariser provably prevents the core from degenerating to a diagonal form (which recovers bilinear fusion).

Experimental Findings

- HRIL outperforms multimodal contrastive baselines on synergy-dominated benchmarks, with gains largest when no pair of modalities alone captures the label.
- Performance scales with Tucker rank $(r_1, \ldots, r_M)$ up to a saturation point, providing a tuneable compute–accuracy trade-off.
- Gains transfer across different encoder architectures and modality combinations.

Ablations and Interpretation

Without synergy regulariser: The core concentrates on diagonals, recovering bilinear-fusion behaviour and losing most of the HRIL advantage.

Tucker rank: Lowering the rank reduces cost with minor accuracy loss; higher ranks plateau beyond a problem-dependent threshold.

Number of modalities: The benefit over pairwise methods grows with M, consistent with the theoretical argument that synergy is a higher-order effect.

Reference

Qun Dai, Zhao Kang et al. **HRIL: Learning Multimodal Synergy via Higher-Order Tensor Modeling**. NeurIPS 2026. arXiv:2610.12393, October 2026.
https://arxiv.org/abs/2610.12393